

Lembar Kerja 1

Kelompok #01

- Ammar Jaddid Habibullah-152024600
- Abdullah Luthfi-152024211

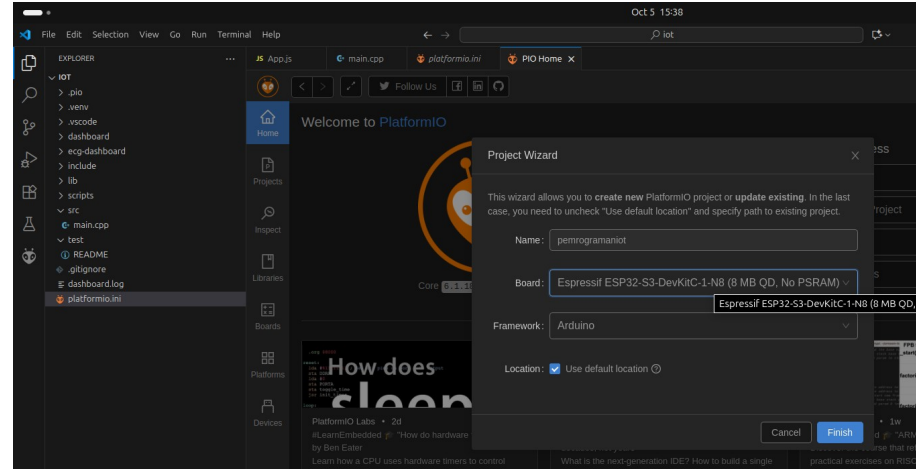
Preparation

- HiveMQ
- ESP32-S3
- Arduino IDE (VS CODE) on your laptop
- Internet connection

Add ESP32 to the board manager

Go to board manager, type “esp32”, and install the two boards that appear in the list.

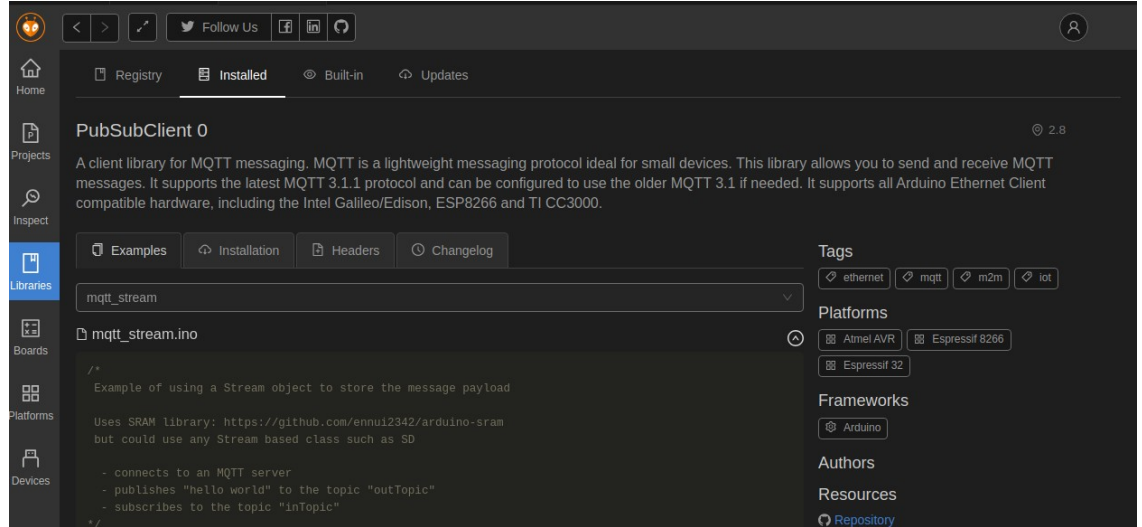
After that, plug the ESP32S3 device, and select **ESP32-S3-Devkitc**.



PubSubClient

Go to library manager, type:
PubSubClient

Install PubSubClient

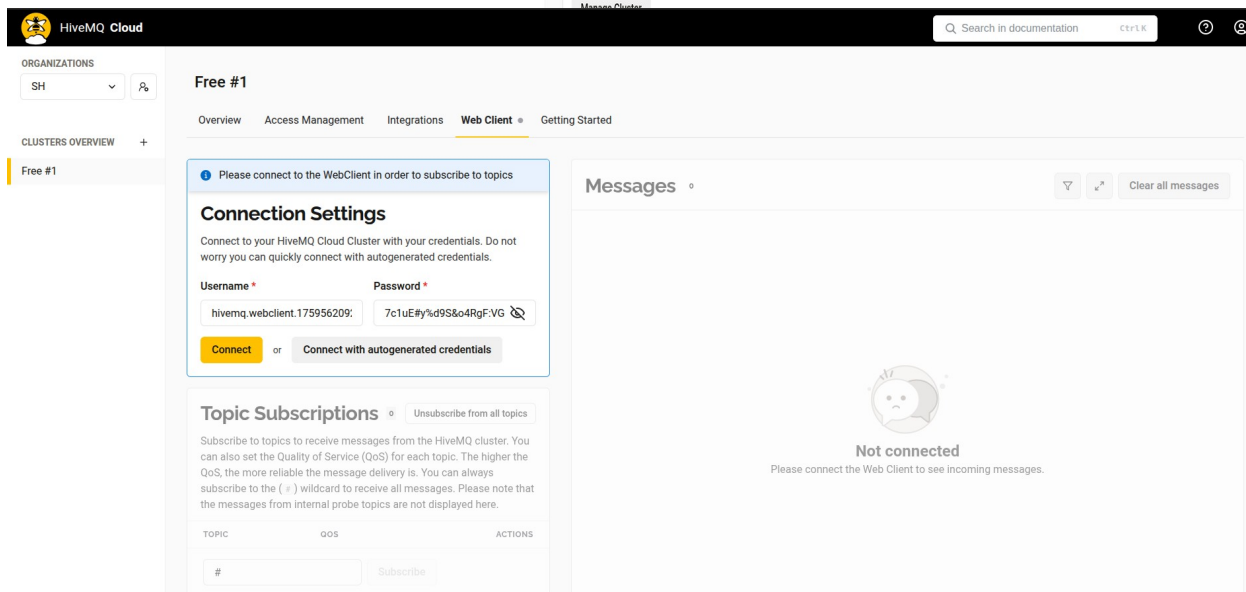
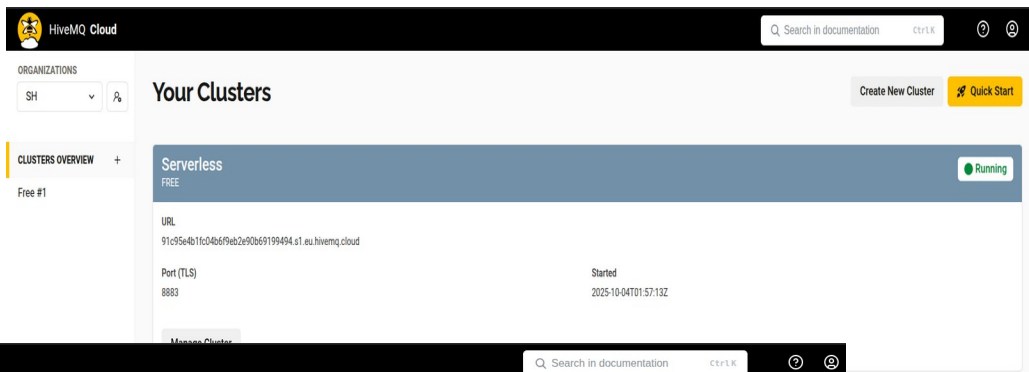


Setup your HiveMQ

Sign-up and login

Create a new cluster

Manage the cluster: add several credentials



HiveMQ address and port

All groups can first try my HiveMQ cluster.

URL : 91c95e4b1fc04b6f9eb2e90b69199494.s1.eu.hivemq.cloud

Port: 8883

NAME	PERMISSION
hivemq.webclient.1759561571639	PUBLISH_SUBSCRIBE
hivemq.webclient.1759562092546	PUBLISH_SUBSCRIBE

“Password” is the same as “Name”

HiveMQ Web Client

Use your own HiveMQ cluster here!

✓ The WebClient is connected

Connection Settings

Connect to your HiveMQ Cloud Cluster with your credentials. Do not worry you can quickly connect with autogenerated credentials.

Username *

hivemq.webclient.175956209:

Password *

.....

Disconnect

Topic Subscriptions

Subscribe to topics to receive messages from the HiveMQ cluster. You can also set the Quality of Service (QoS) for each topic. The higher the QoS, the more reliable the message delivery is. You can always subscribe to the (#) wildcard to receive all messages. Please note that the messages from internal probe topics are not displayed here.

TOPIC	QoS	ACTIONS
🔊 #	QoS: 0	🗑
#	Subscribe	

Messages

0

Topic: ecg/data QoS: 0

{"ecg":-0.200000003,"hr":65,"timestamp":4348286}

1

Topic: ecg/data QoS: 0

{"ecg":-0.050000001,"hr":61,"timestamp":1821491}

2

Topic: ecg/data QoS: 0

{"ecg":0.039999999,"hr":61,"timestamp":1821513}

3

Topic: ecg/data QoS: 0

{"ecg":0.090000004,"hr":61,"timestamp":1821535}

4

Topic: ecg/data QoS: 0

{"ecg":-0.050000001,"hr":61,"timestamp":1821557}

5

Topic: ecg/data QoS: 0

{"ecg":0.029999999,"hr":61,"timestamp":1821579}

6

Topic: ecg/data QoS: 0

{"ecg":-0.07,"hr":61,"timestamp":1821601}

7

Topic: ecg/data QoS: 0

{"ecg":-0.029999999,"hr":61,"timestamp":1821623}

8

Topic: ecg/data QoS: 0

{"ecg":0.100000001,"hr":61,"timestamp":1821645}

9

Topic: ecg/data QoS: 0

{"ecg":0.210000008,"hr":61,"timestamp":1821667}

10

Topic: ecg/data QoS: 0

{"ecg":0.230000004,"hr":61,"timestamp":1821689}

MQTT Publish / Subscribe

```
App.js main.cpp PIO Home .gitignore
src > main.cpp > ...
1 // Include <Arduino.h>
2
3
4
5
6 // WiFi credentials
7 const char* ssid = "gagal";
8 const char* password = "pppppppp";
9
10 // HiveMQ Cloud credentials
11 const char* mqtt_server = "91c95e4b1fc04b6f9eb2e90b69199494.s1.eu.hivemq.cloud";
12 const int mqtt_port = 8883; // Changed from 8884 to standard TLS port
13 const char* mqtt_username = "hivemq.webclient.1759562092546";
14 const char* mqtt_password = "7c1uE#y&d9S&o4RgF:VG";
15
16 // HiveMQ Cloud Certificate
17 const char* root_ca = R"-----BEGIN CERTIFICATE-----
18 MIIFazCCA10gAwIBAgIRAI1Qz7DSQ0NZRGpGu20CiwAwDQYJKoZIhvcNAQELBQAw
19 TzELMAkGA1UEBhMCVVMxKTAnBgNVBAoTIEludGVybmV0IENYIHY3VyaXR5IFJlc2Vh
20 cmNoIEIEdyB3VwMRUwEwYDVQDEwKJUIJHIFjvb3QgWDEwHhcNMTUwNjA0MTEwNDM4
21 WhcNMzUwNjA0MTEwNDM4WjBPMQswCgYDVQDEwKJWUzEpMCcGA1UEChMgS5W0ZXJ1
22 ZXQxQ2VjdXJpdHkgUmVzZWZlY2ggR3JvdXAxFTATBgNVBAMTDEludGVybmV0IENY
23 MTCCAI1wDQYJKoZIhvcNAQEBBQADggIPADCCAgogCggIBAK3oJHP0F0fzm54rVvgc
24 h77ct984kIXuP0ZXoHj3dcKi/vVqbvYATyjb3miGbESTtrFj/RQSa78f0uoxmyF+
25 0TM8ukj13Xnfs7j/EvEhmkvBioZxaUpmZmyPfjxwv60pIgbz5MDmgK7IS4+3mX6U
26 A5/TR5d8mUgjU+g4rk8Kb4Mu0ULXjIB0ttov0DiNewNwIRt18jA8+o+u3dpj+sw
27 T8K0Eut+zwvo/7V3LvSye0rgTBI1DHCNAymg4VMk7BPZ7hm/ELNKjD+Jo2FR3qyH
28 B5T0Y3HsLuJvW5iB4YlcNHlsdu87kGJ55tukmi8mxdAQ47e2RC0FvU396j3x+UC
29 B5iPNgiV5+I3lg02dZ77DnKxHzu8A/LJBdiB3QW0KtZB6awBdpUKD9jfb0SHzUv
30 KBds0pjBqAlkd25HN7r0rFleaJ1/ctaJxQZBKt5ZPt0m9STJEadao0xAH0ahmbWn
31 0LFuhjuefXKnEgV4We0+UXgVcW0PjdAvBbI+e0ocS3MFEVzG6uBQE3x0k3SzyTn
32 jh8BCNAw1FtxNr0HusEwMFXIt4I7mKZ9YIqioymCzLq9gwQb0oMDQaHwBFebwrbw
33 qHyG00aoScQI3Haadr8faqU9GY/rOPnk3sgrDQoo//fb4hVC1CQJ13hef4Y53CI
34 rU7m2Ys6xt0nUW7/vGT1M0NPagMBAAGjQjBAMA4GA1UdDwEB/wQEAWIBBjAPBGNV
35 HRMBaf8EBTADAQH/MB0GA1UdDgQWBRR5tFnme7b15AFzgAiIyBpY9umbbjANBgkq
36 hkiG9w0BAQsFAAOCAgEAVR9YqbyyqFDQDLHYgmkgJyKIRgF1Xipu+ILLaS/V9lZL
37 ubhzEFnTIIZd+50xx+7LSYK05QAvqFyFwhFQDLnrzuB26brJFe+GnY+EGPbk6ZGQ
38 3BebYhtF8GaV0nxvwu077x/Py9auJ/GpsMiu/Xl+mvoiB0v/2X/qk5SisRcOj/KK
39 NfY2PwByVS5uCbMioGziUwthDyC3+6WVw6LLV3xL fHTjuCvjHIInZktHCgK05
40 ORazI4JMPJ+GslWYHb4phowim57iaztX0oJwTdwJx4nLCgdNb0hdjsnvzqvHu7Ur
41 TkXWStAmz0VyyghqpZxjFaH3p03JLF+L+/sKAiuvtd7u+Nxe5AW0wdeRLN8WdC
42 jNPelpzVmbUq4JUagEiuTDkHszxHpFKVK7q4+63SM1N95R1NbDwhscCb+ZAJzVc
43 oyi3B43njT0Q5y0f+1CceWxG1b0V5S2ufpsMlj4Uui0/1lvh+wjChP4kqK0J2qxq
44 4RgqsahDYVvTH9w7jXbyLeiNdd8XM2w9U/t7y0Ff/9y10GE44Za4rF2LN9d1LTPA
45 mRGunUHBcnWvgJBQl9nJEiU0Zsnvgc/ubhPgXRR4Xq37Z0j4r7g15gEEZwxA57d
46 emyPxgcYxn/eR44/KJ4EBs+LVDR3veyJm+kXQ99b21/+jh5Xos1AnX5I1treGcc=
47 -----END CERTIFICATE-----
48
```

Discussions

- Enkripsi: Melindungi data yang dikirimkan antara dua pihak (misalnya, browser dan server web) dari penyadapan.
- Autentikasi: Memverifikasi identitas pihak yang diajak berkomunikasi (biasanya server), memastikan Anda terhubung ke situs yang sah dan bukan situs palsu.
- Integritas Data: Memastikan bahwa data yang dikirim tidak diubah atau dirusak selama perjalanan.

bisa saja iot tanpa tls tapi kemungkinan tidak aman yang sangat tinggi seperti oranglain
bisa melihat data yang sedang dikirim atau yang lebih parah data bisa diubah selama transmisi

Explain TLS Handshake process here:

Proses ini seperti dua pihak yang ingin mengirim surat rahasia. Mereka harus bertemu dulu (handshake) untuk saling memperlihatkan identitas (autentikasi), setuju pada jenis kotak dan gembok yang akan digunakan (cipher suite), dan saling memberikan salinan kunci yang sama (session keys) agar bisa membuka kotak surat tersebut. Setelah itu, barulah mereka bisa saling mengirim surat rahasia dengan aman.